

GIA THIEN-LONG SKYLER DO

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Systems engineer at General Motors specializing in automotive electrical architecture and power management, with a track record of building automation infrastructure that replaces manual engineering workflows. Currently completing an M.Eng. in Systems Engineering and Design at the University of Michigan while working full-time. Background spans mechatronics, robotics, and hands-on fabrication across academic and professional settings.

EDUCATION

University of Michigan, Ann Arbor

M.Eng. Systems Engineering and Design

Concentration: System Architecting and Design

August 2026

University of Illinois at Urbana-Champaign

B.S. Systems Engineering and Design

Minor: Electrical & Computer Engineering

Concentration: Mechatronic System Design

May 2022

PROFESSIONAL EXPERIENCE

General Motors | Low Voltage Mechatronics Infrastructure Power Activation Systems Engineer: Kanban Scrum Lead

Aug 2026 – Present

- Promoted to Scrum Lead for the LVI kanban team, facilitating backlog refinement and PI planning in alignment with SAFe/Agile practices while continuing individual contributor delivery on power architecture specs.
- Drove continuous flow improvements for the team's kanban board, improving throughput visibility across concurrent spec release pipelines.
- Originated "Project Panopticon," an AI-assisted triage system for nightly test failures across two automotive system areas (~100 test scenarios x 20–30 controller builds nightly); reframed the effort as a signal-vs-noise prioritization problem rather than a reporting one to win team alignment.
- Authored the product vision and requirements driving every shipped component, including status visibility, pass/fail transition tracking, root-cause classification, navigable history, and a prioritized daily worklist, which streamlined team standups around a pre-ranked priority list instead of an hour of manual triage.
- Designed and delivered the Deviations feature module end-to-end, replacing a manual spreadsheet for justifying unfixed test failures as a first-class part of the daily board; automated staging of failures still open at cadence close and simplified the data model to significantly cut per-entry effort.
- Integrated the Deviations module with Jira to auto-generate approval tickets with pre-populated fields, links, and evidence, reverse-engineering undocumented field behavior and validating against live Jira before team rollout; shipped as a zero-migration, zero-data-loss upgrade backed by automated tests independent of production data access.
- Partnered with a co-owner on the broader test-reliability platform (overnight automation, cumulative stability views, automated failure analysis with a feedback loop), contributing requirements, design review, and integration decisions.

General Motors | Low Voltage Mechatronics Infrastructure Power Activation Systems Engineer

Sep 2025 – Aug 2026

- Own and deliver three automotive power management architecture specifications (SDV-8280, SDV-8280A, GB-5280) across concurrent cadence release pipelines, covering Partial Network membership, wake mask encoding, and activation/deactivation profile criteria for the GM SDV2 and VIP electrical architectures.
- Designed and implemented a 70-script, 4-gate Python automation platform replacing a fully manual Excel-based workflow; pipeline automates Jira triage, spec validation, Wakeup Data Mask computation, and downstream SANDART ticket creation with Slack notification.
- Architected a markdown-first, git-versioned specification management system for three automotive specs, enabling full change auditability, diff-based review, and automated Excel rendering that replaced opaque binary Excel as the authoritative source.
- Built and deployed a 6-agent AI specialist system (VS Code Copilot) covering the full change-management workflow end-to-end: ticket triage, CCB tracking, spec application, GitHub publishing, criteria outreach, and requirements traceability.
- Conducted proactive criteria outreach across 6+ SME domains (Braking, Energy Management, Vehicle Access, HV Battery, Cabin Thermal, CHM) to resolve blocked activation profiles ahead of architecture milestone deadlines.

- Hosted team AI learning sessions and was selected to represent the LVI organization at the University of Michigan AI Program, translating hands-on automation experience into reusable knowledge for colleagues.

Nagoya University | Research Scholar | Japan-US-Canada
Advanced Collaborative Education Program (JUACEP)

June 2025 – August 2025

Cube Nanosatellite Swarm Simulation

- Designed a dual-path validation strategy for the MAGNARO II cube nanosatellite: a 10cm cubed device scheduled for a 2028 JAXA launch, enabling swarm formation testing to proceed independently of ongoing HILS development.
- Evaluated and selected a full software toolchain (ROS2, PX4, Gazebo, XRCE-DDS, QGroundControl, Micro ROS) for a drone-carried nanosatellite swarm simulation, navigating hardware constraints and protocol tradeoffs between MAVLink and XRCE-DDS and DJI proprietary vs. open-source alternatives.
- Built a multi-drone simulation environment integrating all toolchain components; achieved single-drone keyboard teleoperation, 3-drone simultaneous simulation, and gimbal orientation control for nanosatellite payload positioning.
- Authored system context diagram and signal path documentation mapping all communication paths across ground station, drone, and cubesat agents for both HILS and drone validation branches.
- Produced a structured roadmap for future work spanning three workstreams: multi-drone teleoperation, MAGNARO Micro ROS integration, and magnetorquer-to-drone-orientation physical modeling.

General Motors | TRACK — Technical Rotation and Career
Knowledge

July 2022 – June 2025

TRACK | SDV Architecture Engineer

- Owned requirements for the Partial Network power management strategy of the VIP/SDV2 electrical architecture.
- Evaluated new concepts by analysis or prototyped implementation prior to deployment.
- Wrote requirements and led workgroups for diagnostic and software platform system specifications.
- Developed test cases to improve requirement quality and enable quick feedback for agile implementation iterations.
- Architected transition between power management software infrastructure across vehicle architecture generations.
- Established and maintained working knowledge of Diagnostic and AUTOSAR standards.
- Collaborated cross-functionally to understand and update partial network activation/deactivation criteria.

TRACK | Hardware Platform Architect Engineer

- Supported team-wide goal of managing vehicle fleet/portfolio program complexity, reducing variation and engineering integration overhead resources.
- Interfaced with engineering, marketing, and finance stakeholder teams as part of complexity reduction initiatives.
- Supported GM company-wide pivot toward becoming an automotive technology company by promoting systems-level thinking.
- Supported specific component complexity reduction initiatives to streamline vehicle development.

TRACK | Controls Integration Engineer

- Owned vehicle software integration across 15 mid-size crossover vehicle programs per software release schedule.
- Flashing Vehicle Software Configuration Management (VESCO) archives to development and production vehicles.
- Supported controls manager to ensure DRE compliance with update schedule.
- Completed software release documentation: release notes, flash instructions, validation ID file, hardware PN.
- Updated vehicles for FSIE to conduct DVUL safety testing.
- Supported Movimento (local fleet flashing team) in vehicle update issue resolution.
- Supported root cause analysis for integration issues at the test property.
- Documented and tracked issues with controls manager and domain leads for all development milestones.

COLLEGIATE EXPERIENCE

University of Illinois Urbana-Champaign | Undergraduate
Research Assistant | Autonomous and Unmanned Vehicle Systems
Lab (AUSVL)

Spring 2021 – May 2022

Investigation of Off-Road Vehicle Local Path Planning Alternatives | ISE Research Experience Undergraduate, with Prof. Norris

- Investigated local path planning algorithms for off-road autonomous vehicles in collaboration with Prof. Norris as part of the ISE Research Experience Undergraduate program.
- Implemented the Dynamic Window Algorithm for localized path planning on ROS Jackal robot simulations, evaluating its performance as a candidate for off-road navigation contexts.

Accurate Full State Estimation of a Ballbot Using Optical Flow Sensors | Illinois Scholars Undergraduate Research, HDCL | Sponsored by ISUR & Siemens

- Investigated optical flow sensors as a direct measurement alternative to motor encoders for ball speed estimation on the PURE (Personal Unique Rolling Experience) ballbot developed in the Human Dynamics and Controls Lab.
- Characterized two off-the-shelf optical flow sensors (PMW-3360 and ADNS-3080) by building a custom adjustable focal length testbed with Arduino/Python readout to measure and compare sensor performance.
- Designed and fabricated an iterative bench-top testbed simulating single-axis ball rotation — featuring a laser-cut wheel, bearings, and absolute encoder as ground truth — benchmarking RMS error at speeds from 0.2 to 1 m/s across five sampling rates (50–1000 Hz).
- Selected PMW-3360 at 100 Hz as the optimal sensor based on consistent RMS error across test conditions; ADNS-3080 exhibited unreliable drift despite lower RMS error at higher sampling frequencies.
- Designed CAD sensor mount for the PURE ballbot arrestor leg and developed a dual-measurement validation system combining optical flow and omniwheel encoder with a sinusoidal motor speed test protocol; presented findings at the ISUR symposium.

Illini EV Concept | Shell Eco-Marathon Collegiate Car Competition

Spring 2019 – Spring 2021

- Vice President: led team management, external communications, and overall chapter organization.
- Mechanical Lead: owned design and development of steering, brakes, suspension, drivetrain, and rollcage subsystems.
- Composites Lead: designed and manufactured carbon fiber body panels using in-house CNC foam molds.
- Safety Officer: maintained lab and workplace safety compliance, including COVID protocol enforcement.
- Electronics Team: contributed to battery and motor controller design and assembly.

Other Collegiate Experience

2018 – 2020

- EntreCORPS (Startup Consultant, Fall 2018 – Spring 2020): researched market conditions and developed pricing and market penetration strategies for client startups.
- Illinois Robomasters (Mechanical Engineering Team, 2018 – 2020): designed and fabricated a drone-type combat robot for inter-team competition.

ACADEMIC PROJECTS

CRS Robot Arm Trajectory Control | SE 422: Robot Dynamics and Control | UIUC

- Forward and inverse kinematics of serial manipulators; dynamics and control of actuated robotic systems.
- Practical implementation of programming robotic manipulators using linear control theory.

Self-Balancing Robot | SE 423: Introduction to Mechatronics | UIUC

- Code Composer C programming for TI Microcontrollers.
- DSP and microcontroller usage including SPI, I2C, and PWM concepts.

Double Mass Trolley System Simulation | SE 424: State Space Design for Control | UIUC

- Laplace transforms, state space models, and state feedback design.

ROS/Gazebo Robot Simulations | SE 498: Introduction to Autonomous Vehicle Systems | UIUC

- Simulation methods, localization algorithms, sensor fusion, planning/control, and payload development.

Car Suspension Model Simulation | SE 410: Component Design | UIUC

- Structural member design, material failure principles, design reliability, and analytical simulation.

Product Plan | ISD 520: Introduction to Systems Engineering | University of Michigan

- System definition, requirements, and verification; functional requirements decomposition; problem/solution domain mapping.

CERTIFICATIONS

AI for Engineers *University of Michigan Engineering Professional Education*

- Completed an 8-module applied AI/ML certificate; built and trained models in Python across supervised, unsupervised, and reinforcement learning using automotive, aerospace, manufacturing, and energy case studies.
- Developed hands-on proficiency in generative design, evolutionary algorithms, and topology optimization for AI-driven design and optimization workflows.
- Applied machine learning to autonomous systems (transfer learning, domain adaptation) and predictive maintenance for manufacturing and energy infrastructure.

INCOSE: ASEP *International Council on Systems Engineering — Associate Systems Engineering Professional*

- Earned the ASEP credential by achieving an A in both ISD 520 (Introduction to Systems Engineering) and ISD 521 (Developing and Verifying Requirements) at the University of Michigan — the INCOSE-recognized academic pathway to certification.
- ISD 520 covered systems engineering foundations: stakeholder analysis, requirements decomposition, design reviews, verification planning, model-based SE, and the INCOSE SE Handbook framework.
- ISD 521 covered quantitative requirements development and verification: factor analysis, conjoint/logit regression for stakeholder value modeling, ROC-based test procedure selection, Monte Carlo simulation for programmatic risk, and design of experiments.

Design for Six Sigma: Black Belt *Improving SDV2 Power Management Specifications*

- Identified that the GB-5280 Partial Network specification was difficult to read, test, and trace — causing friction between architects and test engineers and adding cost through repeated clarification cycles.
- Conducted a Voice of Customer survey across the spec consumer community; baseline scores for GB-5280 showed testability at 5.82/10 and traceability at 5.64/10 — establishing quantified targets for improvement.
- Evaluated four specification format concepts (pseudocode, graphics, Excel, Gherkin/BDD feature files) and designed a hybridized solution: Gherkin syntax for behavioral requirement language, Excel for the activation criteria sheet (SDV-8280A), and Word with supporting diagrams for foundational PN content (SDV-8280).
- Post-deployment survey showed precision improved to 7.57/10 and traceability to 7.29/10 versus the GB-5280 baseline — with the split spec architecture now serving as the standard for the power management team.

Design for Six Sigma: Green Belt *Olympian Dynamic Data Dissemination*

- Identified a systemic pain point in GM's Olympian vehicle complexity platform: static, manually maintained datasets with no dynamic update capability and no self-service access for downstream stakeholders including Advanced Controls Managers, Clone Architects, and the Complexity Optimization Group.
- Designed and conducted a structured stakeholder survey across developer and end-user populations; data showed average timeliness satisfaction of 6.94/10 and tool desire score of 8.72/10 — quantifying both the problem and organizational appetite for a solution.
- Recommended a three-tool integrated solution (VCAM + Power BI + PCAT) with supporting process changes, enabling automated HW-to-clone linkage, dynamic data communication, and self-service analysis for enterprise stakeholders.

SKILLS

AI / Automation	MCP Integration, VS Code Copilot Agents, Gated Workflows, Custom Python Pipeline Development
Systems Eng.	SysML, MagicDraw, Rhapsody, Doors Next Generation Agile, V-Model, Requirements Management
CAD / Design	Fusion 360, Inventor, CATIA, SolidWorks, NX PCB: EasyEDA, EAGLE, KiCAD
Fabrication	3D Printing, Machine Shop, Soldering, Woodworking, Metalworking
Programming	Python, C, Java, MATLAB, Simulink, NI LabVIEW, Visual Basic
Robotics / HW	ROS2, Gazebo, PX4, Linux Vehicle Spy 3, INCA Microcontrollers: Arduino, TI MSP430
Tools	Jira, Confluence, SharePoint, MS Office Suite Gherkin BDD
Languages	English (native), Cantonese (fluent), Mandarin (functional), Vietnamese (functional), Japanese (beginner)

LEADERSHIP & ACTIVITIES

Illini Kendo Club

Spring 2020 – May 2022

- President: led club operations, set strategic direction, and oversaw all officer functions.
- Vice President: supported club leadership, coordinated officer team activities.
- Secretary: managed club email account and generated/disseminated promotional and recruitment content.
- Social Chair: planned social events and in-house tournaments.

Triangle Fraternity, Illinois Chapter

Spring 2021 – May 2022

- Vice President: teambuilding, chapter-wide communications, and event management.
- Alumni Relations: collected alumni stories for quarterly newsletter.

Tae Kwon Do Club, UIUC

Spring 2019 – May 2022

- Executive Board Member: Equipment Officer: responsible for purchasing and distributing equipment to club members.

CLAID: Chinese Language and International Development

Spring 2020 – May 2022

- Executive Board Member (Social): planned event logistics and managed club communications.

Martial Arts

Spring 2020 – May 2022

- Tae Kwon Do, Muay Thai.

PROFESSIONAL AND ACADEMIC AFFILIATIONS

- **Triangle Fraternity, UIUC Chapter**
- **Kent School**
- **Avon Old Farms School**